Notes on Noether Charge Decay in Learning Dynamics: Learning as Symmetry Breaking

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Abstract

We view the training of a deep network as a **dissipative dynamical system** and study the behavior, during learning, of the Noether charge $C = W_1 W_1^\top - W_2^\top W_2$ associated with the rescaling symmetry. Under pure gradient flow C is conserved (Du–Arora balancedness); once weight decay breaks the symmetry, C decays exponentially as $\dot{C} = -2\mu C$. This mechanism is already present in prior work (notably Kunin et al.'s *Neural Mechanics*), and **is not the contribution of this paper**. Our thesis is one layer above it: *a direction in which the charge is conserved is exactly a direction in which the network learns nothing.* We give two mutually reinforcing results. (i) Under isotropic, constant weight decay the decay rate is set solely by the regularization strength and is **independent of the task** (a negative result), so the "decay rate" as an observable carries no learning information. (ii) The charge measures *how* the network organizes itself internally, which is **orthogonal** to *what* task it learns (the singular spectrum of the product $W_2 W_1$); this orthogonality is established through a negative experiment (the attempt to read the task off the charge fails, and the manner of failure is exactly what proves the orthogonality). We then describe learning as phase-space contraction onto a low-dimensional attractor, and distill two invariants meant to survive from the idealized regime to realistic SGD: a **symmetry-breaking number** (an integer) and an **attractor dimension**. Everything is demonstrated on one two-layer linear toy model, with explicit labeling of what is rigorous, what is illustrative, and what is still aspirational.

1 Introduction

The only stable conserved structure in learning comes from the reparameterization redundancy of the network. **TODO: prose.**

Boundary with prior work (critical). Neither the bare decay law $C(t) = C(0)e^{-2\mu t}$ nor balancedness conservation is new here: the former appears in Kunin et al.'s *Neural Mechanics* [1] (which treats broken conservation as an instrument for tracking SGD trajectories), and the latter in Du et al. [2] and Arora et al. [3]. Related are Tanaka–Kunin's Noether learning dynamics [4] and Ziyin et al.'s noise equilibrium [5]. **We do not re-derive this mechanism**; the contribution is the interpretive/classification layer on top of it — conservation as a learning blind spot, how⊥what, and "the decay rate carries no learning information, hence the turn to discrete invariants." **TODO: 2–3 sentences of positioning.**

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2 Rescaling Symmetry and the Conserved Charge C

2.1 A conserved quantity from symmetry

For a two-layer network $\hat{y} = W_2 W_1 x$ and any invertible G, the change $W_1 \to GW_1,\ W_2 \to W_2 G^{-1}$ leaves the product $W_2 W_1$, hence the loss L, invariant. The corresponding conserved matrix quantity is

$$\boxed{C = W_1 W_1^\top - W_2^\top W_2.} \tag{1}$$

Proposition 2.1 (Conservation of C under gradient flow; Du–Arora balancedness)**.** *Since L depends on the parameters only through $P = W_2 W_1$, we have $\partial L/\partial W_1 = W_2^\top G_P$ and $\partial L/\partial W_2 = G_P W_1^\top$ with $G_P = \partial L/\partial P$. Under gradient flow $\dot{W}_i = -\partial L/\partial W_i$,*

$$\dot{C} = \dot{W}_1 W_1^\top + W_1 \dot{W}_1^\top - \dot{W}_2^\top W_2 - W_2^\top \dot{W}_2 = 0,$$

the four terms cancelling pairwise.

Remark 2.2 (Noether / moment-map aside, load-bearing nowhere)**.** *The conservation of C is exactly the algebra of Proposition 2.1; it does not rely on a Lagrangian, an action, or a balance law. Intuitively it corresponds to "rescaling is a continuous symmetry of the loss, and symmetry implies conservation" (a gradient-flow Noether / moment map), but we do not build any theorem on that intuition.*

2.2 Breaking the symmetry: weight decay makes C decay

Weight decay adds $-\mu w$ to the gradient (equivalently $\frac{\mu}{2}\|w\|^2$ to the loss), a term not invariant under rescaling. Substituting into the flow,

$$\dot{C} = -2\mu(W_1 W_1^\top - W_2^\top W_2) = -2\mu\, C \quad \Longrightarrow \quad C(t) = C(0)\, e^{-2\mu t}. \tag{2}$$

The decay rate is 2μ, purely first order. This law already appears in prior work [1] and serves here as the starting point, not the contribution. The toy model fits a rate $0.6008 \approx 2\mu$ (§7); we further verified that velocity damping $-\mu\dot{w}$ does *not* make C decay, so the decay is genuinely due to the symmetry-breaking weight-decay term.

3 The Idealized System and a Three-Tier Ladder

3.1 The idealized system: constant loss

A flat loss ⇒ no gradient, no force ⇒ all Noether charges conserved, with the number of symmetry directions equal to the full dimension N. But precisely for that reason nothing is learned. This gives the core inversion that runs through the paper:

> **Perfect conservation = perfect "not learning".** Conservation is not an achievement of learning but its absence.

3.2 Three tiers

Real SGD sits at Tier 3, with three things Tier 2 lacks: noise (minibatching), anisotropy (especially with Adam), and time-varying μ (warmup, learning-rate decay). We establish theorems only at Tier 2; Tier 3 is a direction, not a result.

Table 1: A three-tier idealization from ideal to realistic. This paper's main arena is Tier 2.

Tier	**Loss**	**Reg./dissipation**	μ	C **behavior**
Tier 1	constant (flat)	none	—	conserved (no learning)
Tier 2 (main arena)	sloped	isotropic weight decay	constant	exp. decay, rate 2μ
Tier 3	sloped	anisotropic + noise	time-varying	drift/fluctuation (real SGD)

4 Main Results: From a Negative Result to an Orthogonal Decomposition

Theorem 4.1 (Learning = symmetry breaking, C decays)**.** *Under pure gradient flow $\dot{C} = 0$ (symmetry unbroken; that direction is a blind spot, not learned); once a regularizer breaks the symmetry, $\dot{C} = -2\mu C$, and $\dot{C} = 0 \iff$ that direction does not participate in learning.*

Theorem 4.2 (Negative result: the isotropic decay rate is task-independent)**.** *Under isotropic, constant weight decay, $C(t) = C(0)e^{-2\mu t}$, with a rate fixed solely by the regularization strength 2μ and independent of task, data, or content. Hence the "decay rate" as an observable carries no learning information.*

Theorem 4.3 (Orthogonal decomposition: how $\perp$ what)**.** *Learning splits the parameters into two orthogonal parts:*

$$\underbrace{\textit{task content (what)}}_{\textit{singular spectrum of } W_2W_1} \perp \underbrace{\textit{internal organization (how)}}_{\textit{Noether charge } C}.$$

The charge measures how *the network learned (internal cross-layer organization / gauge freedom), orthogonal to and independent of* what *task it learned: two networks learning the same task can have different charges, and networks learning different tasks can have the same charge.*

Establishing the orthogonality through a negative result. The attempt to "tell tasks apart from the charge decay" *fails*: across tasks of different rank, the entire Noether spectrum decays to zero under weight decay, independent of the task. The manner of failure is precisely what proves the orthogonality: by contrast, the singular spectrum of the product W_2W_1 varies clearly with task rank (§7, Fig. 1). The task information (what) lives in the product spectrum; the charge (how) is orthogonal to it. **TODO: add the structural argument — after quotienting the parameter manifold by GL, what lives entirely in the product, and C is the coordinate along the symmetry orbit (moment map).**

Task information can only enter through anisotropy. For the charge decay to carry task information, the force acting on the symmetry directions must be *anisotropic* (a non-Euclidean metric or anisotropic noise). Under isotropic force the decay is orthogonal to the task (this is the source of Theorem 4.2); only anisotropy (natural gradient, Adam, anisotropic noise) lets the decay spectrum carry task information. **TODO: this is the key to Tier 3; needs a quantitative experiment (§8).**

5 Dissipative Layer: Phase-Space Contraction and the Attractor

Theorem 5.1 (Contraction onto an attractor)**.** *Learning (dissipation) monotonically contracts the phase-space volume, and the motion settles onto an attractor of lower dimension than the parameter* *space (the dissipative version of Liouville's theorem: conservative = volume preserved, dissipative = volume contracts).*

Theorem 5.2 (Contraction rate = sum of charge decay rates = sum of Lyapunov exponents)**.** *The three are equal. In the toy model $\|C\| \sim e^{-2\mu t}$ (C is quadratic, rate 2μ), matching theory exactly (Fig. 1).*

Remark 5.3 (Aspirational: contraction rate = entropy production rate)**.** *The contraction rate of a dissipative system equals its entropy production rate, so learning can be seen as an irreversible entropy-increasing process (a "second law" for learning). Listed as an outlook.*

6 Two Core Invariants

Following the shared trait of Poincaré (topological invariants), Morse (critical-point indices), and dissipative systems (attractor dimension) — capture invariants that stay stable and classifiable above the unsolvable complexity, rather than rates the optimizer locks — we distill two quantities (*a program/outlook here, not a proven theorem*): the **symmetry-breaking number** (an integer, a topological invariant): how many originally conserved (dead) directions learning has "activated" = the depth of learning; and the **attractor dimension** (a dimension invariant): how many dimensions the system actually lives in after learning = the effective dimension. They are complementary, with

$$\text{symmetry-breaking number} \;+\; \text{attractor dimension} \;\approx\; N. \tag{3}$$

The decay rate fluctuates and does not generalize; these two integer/dimension invariants are meant to survive from Tier 2 to Tier 3. **TODO: give the symmetry-breaking number a rigorous definition (e.g. the effective rank of the force on the symmetry directions / the count of nonzero Hessian spectrum along gauge directions), and test its stability at Tier 3.**

7 Toy Model Verification

All results are demonstrated on one two-layer linear net $\hat{y} = W_2 W_1 x$, $C = W_1 W_1^\top - W_2^\top W_2$.

Honest boundary. Hard numerical verification: Theorems 4.1, 4.2, 5.2, 4.3 (of which 5.2 matches the theoretical $e^{-2\mu t}$ exactly, the hardest). **Illustrative / to be made rigorous**: the anisotropy effect (weak in the toy model, but it does vary with task). **Aspirational**: entropy production, and the stability of the two invariants at Tier 3. Scope: the above hold rigorously at Tier 2 (isotropic, constant μ); Tier 3 is a direction and must not be extrapolated to directly.

8 Discussion and Outlook

A A Road Not Taken: the Second-Order Lagrangian Framing and Why We Drop It

One can, if one insists, dress this dissipation in a "mechanical" shell: introduce an inertia m, write training as a second-order Lagrangian $L = \frac{1}{2}m\,\|\dot{w}\|^2 - L_{\text{loss}}$, and apply a dissipative Noether theorem (balance law $dQ_N/dt = \langle F, \delta w\rangle$) to obtain a velocity-bearing momentum charge $Q_N = m\langle \dot{w}, \delta w\rangle$ with rate μ/m. **We drop this route** because: (i) Q_N and C are *two different charges* with mismatched rates (μ/m vs 2μ); (ii) introducing m only to take $m \to 0$ back to first order is a reconstruction, not a new result; (iii) the dissipative Noether law $dQ_N/dt = -\lambda Q_N$ is already covered by standard

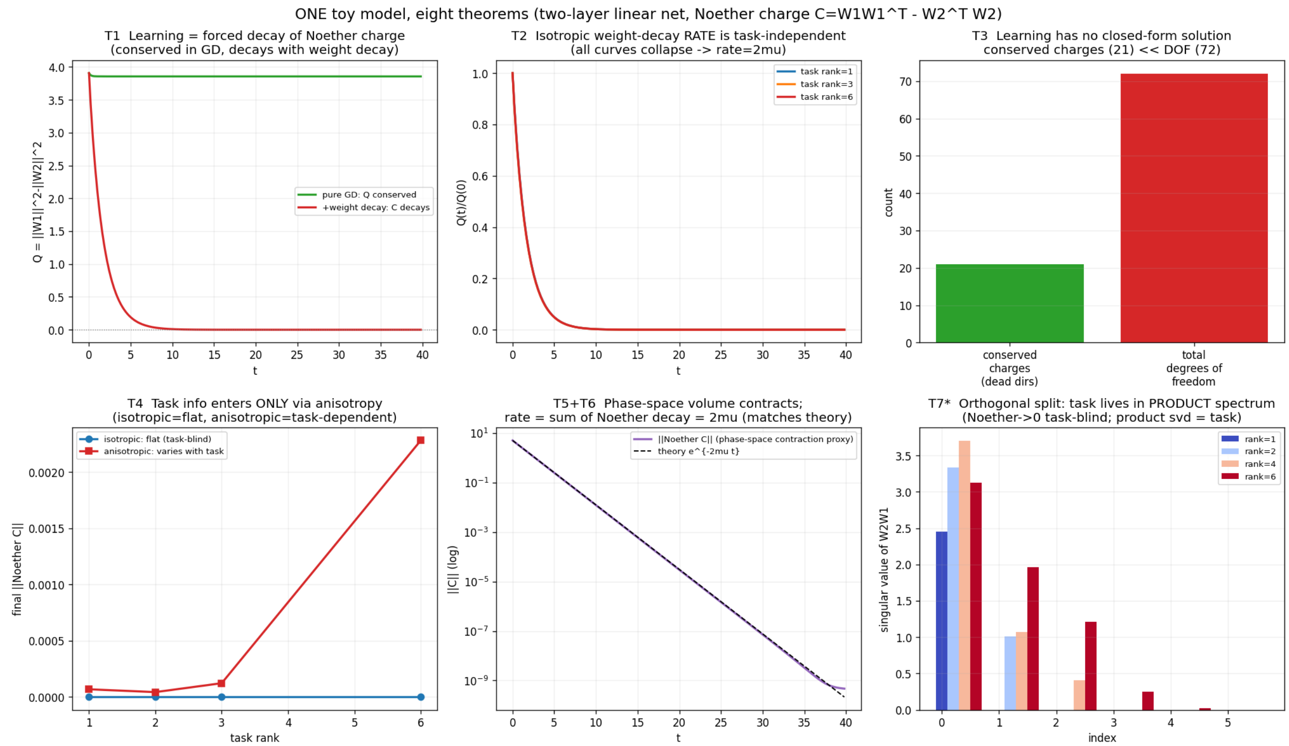

Figure 1: One toy model demonstrating the main results ($d_{\text{in}} = d_h = d_{\text{out}} = 6$, 72 DOF, the charge a 6×6 symmetric matrix with at most 21 symmetry directions).

treatments of damped systems [6]. Hence m, Q_N, and the balance law are kept out of the main text and confined to this note.

References

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